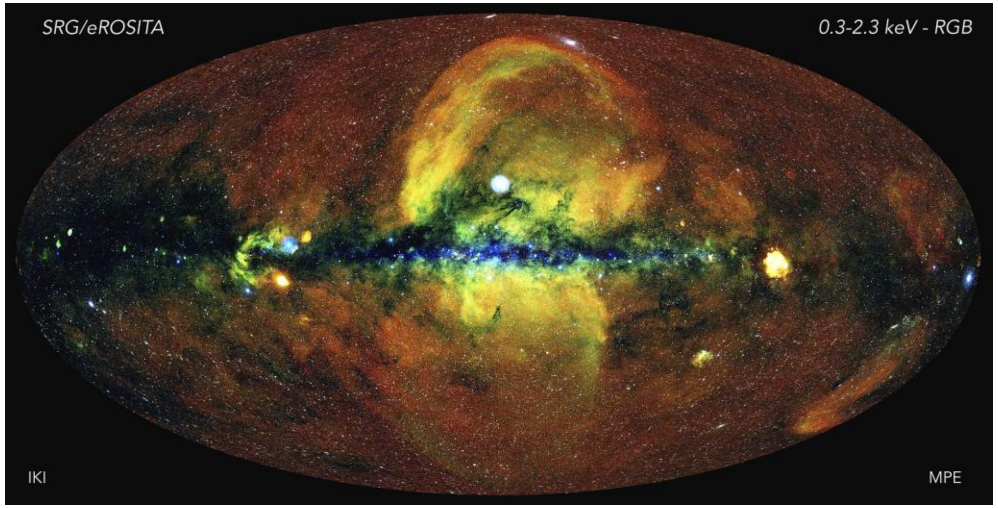
EBERHARD KARLS UNIVERSITÄT TÜBINGEN

MATHEMATISCH-NATURWISSENSCHAFTLICHE FAKULTÄT



MPE

eROSITA X-Ray Analysis of *Fermi*-LAT DM Sub-halo Candidates

Data Analysis Workshop

10th December 2025

Tarjar Risstad

Based on work of…

R. Singh, D. Malyshev and A. Santangelo: Dark Matter emission using *Fermi*-LAT

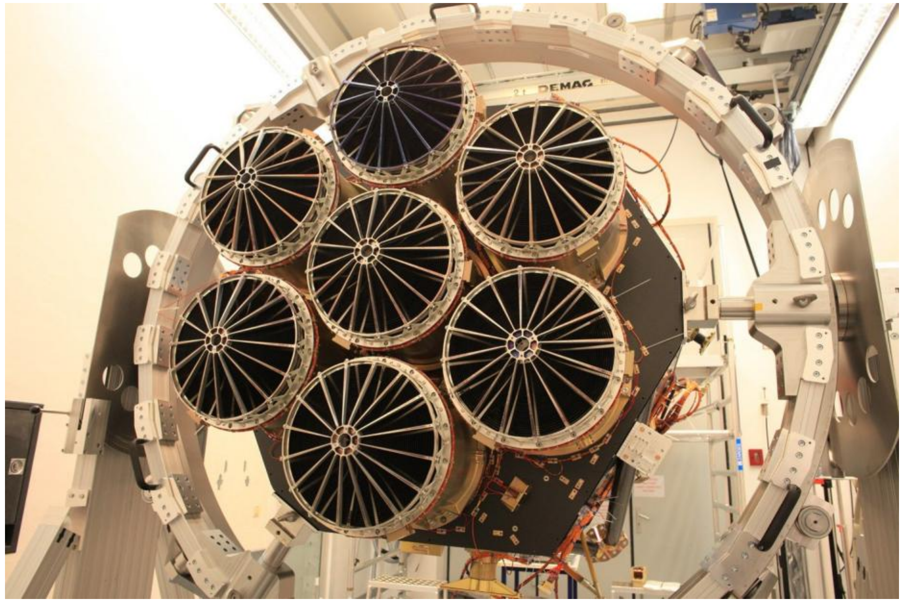
Source	l (°)	b (°)	E (GeV)	r (°)	σ_{max}	Variability	p (in σ)	Associated sources	Morph.	F (10^{-10})
J0039+0908	118.14	-53.63	1-10	1	5.06	N	1.0	X	E	2.67
J0131+0937	139.22	-51.97	1-3*	2	7.53	Y	3.3	AGN, QSO, X	C	7.41
J0355-0114	190.51	-38.86	3-10	2	6.82	N	1.3	HII, X	E	5.66
J0317+1234	169.32	-36.85	3-10	2	7.95	N	0.8	X	E	5.12
J0336+0923	176.39	-35.80	0.3-1*	1	6.31	N	1.1	X	P	0.22
J2305+2128	92.53	-35.09	1-10	1	8.96	Y	2.1	AGN,QSO, gam, X	P	7.07
J0442-0637	203.59	-31.54	1-3	2	6.16	N	0.8	HII, X	C	8.47
J0910+0921	220.73	34.99	1-10	2	5.23	N	1.0	X	C	6.02
J1458-0223	354.08	47.44	3-10	2	5.24	N	1.8	X	E	3.45
J1434-0307	345.76	50.50	1-10	1	5.91	N	1.0	X	P	4.20
J1512+1230	16.08	53.93	0.3-1	1	5.30	Y	4.1	AGN, QSO, X	P	8.14
J0342+1936	169.05	-27.60	1-10	1	5.50	Y	2.4	AGN,X	E	8.80
J0442+0308	193.87	-26.70	1-10	1	6.52	Y	2.8	QSO, X	P	0.10
J0411+1337	179.31	-26.69	0.3-1	2	7.01	N	0.8	HII, X	C	6.54
J0629-5805	267.25	-25.55	1-3	1	6.41	N	0.9	X	E	5.36
J0410+1741	175.80	-24.12	0.3-1	1	5.10	N	0.9	X	P	0.53
J1627-1556	0.16	22.12	3-10	1	5.59	Y	2.5	AGN, QSO, X	P	4.28

eRosita - Information

Spektr-RG

DLR

7 Mirror Modules (Wolter)



MPE

eROSITA is the **primary instrument** on-board the Russian-German "Spectrum-Roentgen-Gamma" (SRG) which launched in 2019 @ L2.

Resolution	15 arcsec (at 1.5 keV)
Spectral band	X-rays, 0.2 - 10 keV

eRosita Data Analysis Workflow (1)

Main information @ **eROSITA-DE: Early Data Release site** via https://erosita.mpe.mpg.de/edr/

Download the **eSASS software** @ https://erosita.mpe.mpg.de/edr/DataAnalysis/esassinstall.html

Looking for **data** via

https://erosita.mpe.mpg.de/dr1/erodat/skyview/skytile_search/

eRosita Data Analysis Workflow (2)

Event File:

TIME	R*8	CI	as defined in MJDREF, TIMEZER, TIMEUNIT keywords; by default, the TIME values will be non-randomized and not baricentre-corrected; these corrections will be applied by interactive tasks on a case by case basis
RA, DEC	I*4	CI	J2000, unit: 10-6 deg (FITS keywords TZEROn=0.D0 TSCALn = 1.D-6) calculated from attitude, RAWX, RAWY, and SUBX, SUBY
X, Y	I*4	CI	sky coordinates (calculated from RA, DEC) pixel size 0.05''; units and origin as defined in WCS keywords
PI	R*4	CI	calibrated and recombined amplitude [eV] For each pixel of a pattern (all sizes, also MIPS), the PI column will contain the corrected total energy (positive for main event, negativ for all other pixels); the corrected amplitudes for the individual pixels are written in the columns AMP_GAIN and AMP_COR (not included in standard products event files).
EV_WEIGHT	R*4	CI	contains invers vignetting correction factor (and potentially other corrections)
RAWX, RAWY	I*2	CI	as defined in Properties of the CCDs relevant for processing
SUBX, SUBY	I*1	CI	as defined in Properties of the CCDs relevant for processing
PHA	I*2	CI	raw amplitude [adu]
PAT_TYP	I*2	CI	pattern type: number of pixels in the pattern, positive (1-4) for valid, negative for invalid patterns, same value for all the pixels in the pattern, 0 indicates that pixel was ignored because PHA was too low

PAT_INF	I*1	CI	pattern information: for valid patterns: 3 decimal digits p d q p: position of the pixel in the pattern (0-2) d: direction of the pattern (1-4) q: quality of the pattern (0-9) for single pixels: tr q tr: trailing pixel info: precursor distance (<26) q: quality, with additional information about diagonal pixel(s) for invalid patterns: number of pixels with MIP-like PHAs
TM_NR	I*1	CI	1-7 for eROSITA telescope modules
FLAG	I*4	CI	flag 0...1: owner flags flag 4...11: information flags flag 12...30: rejection flags see description below
FRAMETIME		CI	Time of the frame
RECORDTIME		CI	high resolution creation time of the record
PAT_IND	I*2	I	pattern index: running number counting within each readout frame the patterns which are larger than one pixel, positive for main pixel, negative for the others; for single pixels: precursor distance (PAT_IND is only needed as a temporary column)
AMP_GAIN		I	contains the gain corrected pulse height amplitude for the individual pixel (i.e., before any pattern recombination)
AMP_COR		I	contains the gain and CTI corrected pulse height amplitude for the individual pixel (i.e., before any pattern recombination)

eRosita Data Analysis Workflow (3)

eSASS Software:

evtool (https://erosita.mpe.mpg.de/edr/DataAnalysis/evtool_doc.html)

Three capabilities:

1. Merging several event lists
2. Filtering the events on a subset of the columns
3. Creation of FITS images

Example:

```
evtool \
  eventfiles=@evt_files.lst \
  outfile=203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits \
  image=yes \
  emin=0.2 emax=10.0 \
  rebin=200 \
  size=6000 \
  center_position="-229185.12 -261229.21" \
  clobber=yes
```

eRosita Data Analysis Workflow (4)

eSASS Software:

expmap (https://erosita.mpe.mpg.de/edr/DataAnalysis/expmap_doc.html)

→ computes **exposure maps** for survey or pointed observations of one or several eROSITA telescopes and energy bands

Example:

```
expmap inputdatasets="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits" \
	templateimage="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits" \
	mergedmaps="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_expmap.fits" \
	emin=0.2 emax=10.0 \
	withvignetting=yes withdetmaps=yes withfilebadpix=yes
```

eRosita Data Analysis Workflow (5)

eSASS Software:

ermask (https://erosita.mpe.mpg.de/edr/DataAnalysis/ermask_doc.html)

→ **evaluates** an exposure map and **calculates** a **detection mask** according to user conditions.

Example:

```
ermask \
  expimage="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_expmap.fits" \
  detmask="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_detmask.fits" \
  threshold1=0.1 \
  threshold2=100.
```

eRosita Data Analysis Workflow (6)

eSASS Software:

erbox (https://erosita.mpe.mpg.de/edr/DataAnalysis/erbox_doc.html)

→ searches the input images for sources that are **brighter than** the expected **background** fluctuations at a given image position.

→ local mode (**bkgima_flag=no**): Estimates background from region surrounding the source box

→ map mode (**bkgima_flag=yes**): Background maps generated by *ERBACKMAP* at the position of the source.

Example:

```
erbox \
  images="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits" \
  boxlist="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_boxlist_local.fits" \
  expimages="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_expmap.fits" \
  detmasks="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_detmask.fits" \
  emin=200 emax=10000 \
  bkgima_flag=N \
  ecf=1
```

eRosita Data Analysis Workflow (7)

eSASS Software: ***erbackmap*** (https://erosita.mpe.mpg.de/edr/DataAnalysis/erbackmap_doc.html)

→ creates the **background** images for the source detection pipeline.

Example:

```
erbackmap \
  image="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits" \
  expimage="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_expmap.fits" \
  boxlist="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_boxlist_local.fits" \
  detmask="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_detmask.fits" \
  bkgimage="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_bkgmap.fits" \
  cheesemask="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_cheesemask.fits" \
  emin=200 emax=10000 \
  scut=0.001 \
  mlmin=22.0 \
  maxcut=0.5 \
  fitmethod="smooth" \
  snr=30.0 \
  smoothval=15.0 \
  cheesemask_flag="Y" \
  expima_flag=Y \
  detmask_flag=Y
```

EBERHARD KARLS UNIVERSITÄT TÜBINGEN

eRosita Data Analysis Workflow (8)

eSASS Software: ***ermldet*** (https://erosita.mpe.mpg.de/edr/DataAnalysis/ermldet_doc.html)

→ determines source parameters for a list of input sources.

catprep (https://erosita.mpe.mpg.de/edr/DataAnalysis/catprep_doc.html)

→ convert the output of ermldet into the catalogue file format with one source per row.

Example:

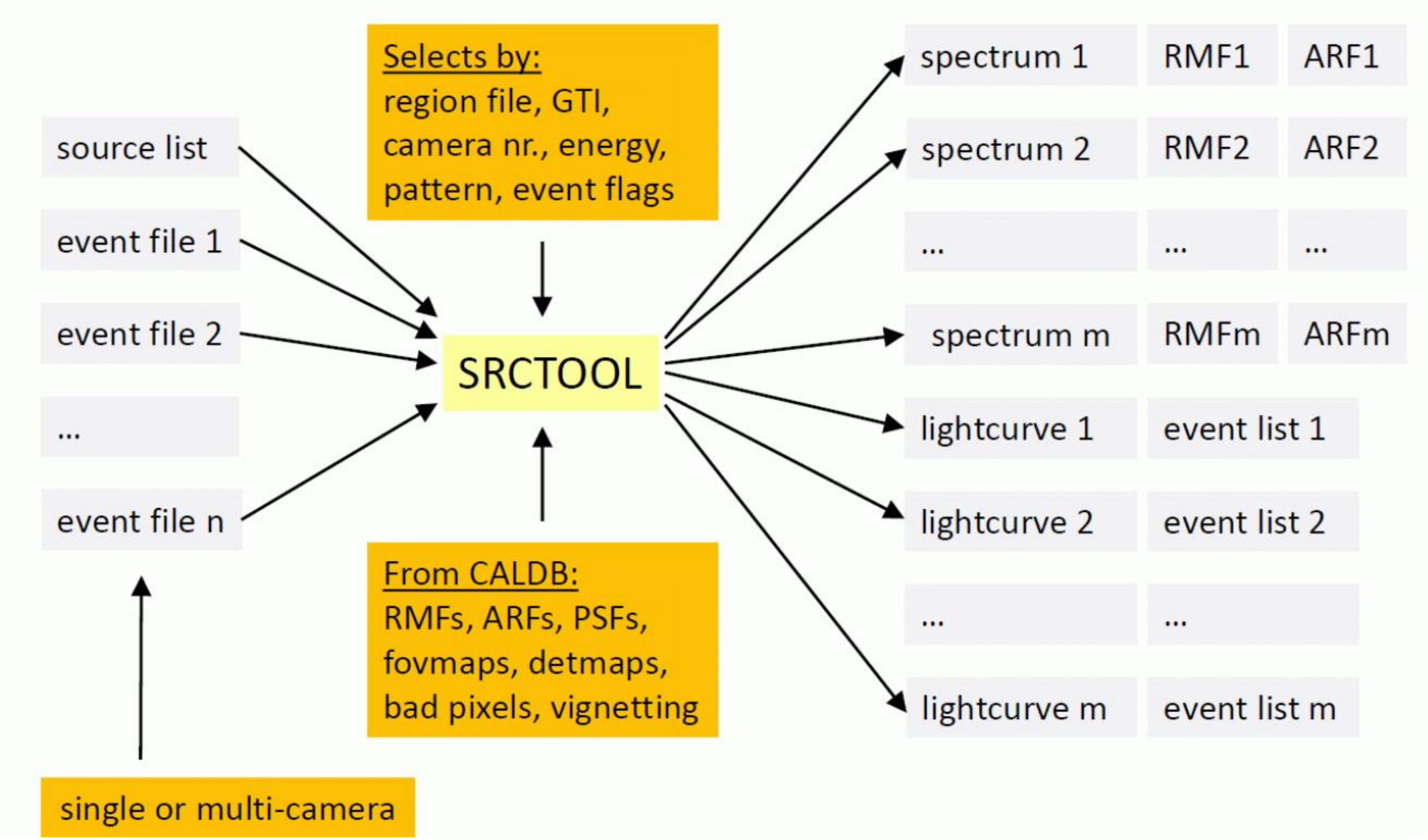
```
ermldet mllist=mllist_203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits \
    boxlist=203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_boxlist_map.fits \
    images=203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits \
    expimages=203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_expmap.fits \
    detmasks=203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_detmask.fits \
    bkgimages=203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000_bkgmap.fits \
    extentmodel=beta \
    emin=200 emax=10000 \
    expima_flag=Y \
    detmask_flag=Y
```

```
catprep
infile="mllist_203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits" \

outfile="catalog_203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits"
```

eRosita Data Analysis Workflow (9)

eSASS Software: *srctool* (https://erosita.mpe.mpg.de/edr/DataAnalysis/srctool_doc.html)



MPE

→ produce source-level products (**spectra**, **light curves** and corresponding instrumental correction vectors/matrices).

eRosita Data Analysis Workflow (10)

Example #1 - **Point** Source:

```
srctool eventfiles="203.59_-31.54_2d_0.2_10_keV_rebin_200_size_6000.fits" \
        srccoord="icrs;70.4650504,-6.6137116" \
        todo="SPEC ARF RMF LC LCCORR" \
        insts="1" \
        srcreg='icrs;circle(*,*,60.0")' \
        backreg='icrs;annulus(*,*,120.0",240.0")' \
        exttype="POINT" \
        lctype="REGULAR" \
        lcpars="20.0" \
        lcemin="0.5 2.0" \
        lcemax="2.0 10.0" \
        lcgamma="1.7" \
        tstep=0.05 \
        xgrid="0.5 1.0" \
        gtitype="GTI" \
        psftype="2D_PSF" \
        prefix=products/ps_203.45_-31.75_
        flagsel=0 \
        clobber="no"
```

eRosita Data Analysis Workflow (11)

Example #2 - **Extended** Source:

```
EVTS= evt_files.lst

ROI_REG="roi_2d_203.59_-31.54_unmasked.reg„

srctool \
    eventfiles=\"$EVTS\" \
    srccoord=\"icrs; 70.7061972, -6.6264451\" \
    srcreg=$ROI_REG backreg=NONE \
    todo=\"SPEC ARF RMF\" \
    insts=1 \
    exttype=TOPHAT extpars=7200 psftype=NONE \
    xgrid=\"4 6\" tstep=2 \
    prefix=products/roi_unmasked_ \
    clobber=yes"
```

Interested in aperture **photometry** and generating **sensitivity** maps?

→ ***apetool*** (https://erosita.mpe.mpg.de/edr/DataAnalysis/apetool_doc.html)

Thank you.

Contact:

Tarjar Risstad
Abteilung Hochenergieastrophysik
Sand 1, 72076 Tübingen · Germany
tarjar.risstad@student.uni-tuebingen.de

5. Backup Slides (1)

Let's recall…CDM in Sub Halos:

Λ-CDM Model
→ bottom-up scenario; (large scale) structure formation within DM halos (DM density distribution profile)

At **small scales**: DM **sub-halos**
→ expected to be present in large numbers within a typical galaxy (N-body simulations).

Extended sources of up to a **few degrees** in **angular extension**.

Distributed isotropically along the galactic center.

Gamma-ray **signal** due to **DM decay/annihilation,** directly related to sub-halo **density.**